

A Theory of Generalized Claims

A Stochastic Routing and Queueing Framework for Institutional Systems

Ananya Sen and Soumadeep Ghosh with
ChatGPT (OpenAI)

Kolkata, India

Abstract

We develop a stochastic network model of institutional systems in which heterogeneous social, economic, and political interactions are represented as generalized claims processed through capacity-constrained institutional channels. Claims are routed endogenously by agents based on congestion-sensitive payoffs, generating coupled queueing dynamics. We derive stability conditions, establish a mean-field limit, characterize regime transitions in congestion, and quantify inefficiency relative to a planner benchmark. We further provide a dual representation of institutional congestion in terms of shadow prices associated with capacity constraints.

The paper ends with “The End”

1 Introduction

Institutional systems process heterogeneous interactions such as disputes, requests, and claims. These interactions must be allocated to institutional mechanisms with finite capacity, generating congestion effects.

We model such systems as stochastic networks in which agents route generalized claims through institutional channels. Channels differ in quality and capacity, and their performance depends on congestion.

We restrict attention to systems where institutional capacity is finite and congestion is observable through backlog accumulation.

Remark 1 (Interpretation guide). *At each time period, agents submit claims which are routed to institutional channels. Each channel processes claims subject to capacity constraints, generating congestion. Agents respond to congestion by adjusting routing decisions. The system evolves as a coupled stochastic network of queues.*

2 Model

2.1 Claims

Definition 1 (Claim). *A claim is a tuple*

$$q = (a, x, x', \rho),$$

where a is an agent, x is a current state, x' is a demanded state, and ρ is a justification rule.

2.2 Institutional Channels

Definition 2 (Channel). *Each channel $\kappa \in \mathcal{K}$ is characterized by*

$$(\theta_t(\kappa), \alpha_t(\kappa), R_t(\kappa)),$$

where θ_t is quality, α_t is authority, and R_t is backlog.

2.3 Routing

Definition 3 (Routing rule). *Agents choose channels according to:*

$$\kappa^*(q) \in \arg \max_{\kappa \in \mathcal{K}(q)} U(q, \kappa; R_t(\kappa)).$$

3 Institutional Dynamics

Assumption 1 (Service regularity). *For each channel κ , service output satisfies:*

$$\mu_t(\kappa) = g(\theta_t(\kappa), R_t(\kappa), \xi_t),$$

where ξ_t is i.i.d. noise and g is Lipschitz in R_t and non-decreasing in θ_t .

Assumption 2 (Welfare structure). *Social cost is additive in congestion:*

$$\mathcal{W}_t = \sum_{\kappa \in \mathcal{K}} w(R_t(\kappa)) + c(\theta_t(\kappa), \alpha_t(\kappa)),$$

where $w' > 0$, $w'' \geq 0$.

3.1 Queue evolution

$$R_{t+1}(\kappa) = R_t(\kappa) + A_t(\kappa) - \mu_t(\kappa)$$

3.2 Capacity constraint

$$\mu_t(\kappa) \leq \theta_t(\kappa) \bar{C}(\kappa)$$

3.3 Congestion feedback

$$\frac{\partial U}{\partial R_t(\kappa)} < 0$$

4 Queueing Reduction

Theorem 1 (Singleton-channel reduction). *If $|\mathcal{K}| = 1$, the system reduces to:*

$$R_{t+1} = R_t + A_t - \mu_t,$$

a stochastic queueing process.

Proposition 1 (Stability condition). *A sufficient condition for stability is:*

$$\mathbb{E}[A_t] < \mathbb{E}[\mu_t].$$

5 Routing Network Equilibrium

Definition 4 (Network system). *For $|\mathcal{K}| > 1$, channels form a coupled queueing network with endogenous routing.*

Proposition 2 (Feedback effect). *Congestion induces rerouting:*

$$R_t(\kappa) \uparrow \Rightarrow A_t(\kappa) \downarrow.$$

6 Mean-Field Limit

Definition 5. *Let μ_t^n denote the empirical distribution of channel states.*

Theorem 2 (Mean-field convergence). *As $n \rightarrow \infty$, μ_t^n converges weakly to μ_t satisfying:*

$$\partial_t \mu_t = \mathcal{F}(\mu_t).$$

Theorem 3 (Phase transition). *There exists λ_c such that:*

$$\lambda < \lambda_c \Rightarrow \text{stable}, \quad \lambda > \lambda_c \Rightarrow \text{divergent backlog}.$$

7 Welfare and Inefficiency

Definition 6 (Welfare loss).

$$\mathcal{W}_t = \sum_{\kappa \in \mathcal{K}} w(R_t(\kappa)) + c(\theta_t(\kappa), \alpha_t(\kappa)).$$

Definition 7 (Inefficiency gap).

$$\Gamma_t = \mathcal{W}_t^{\text{dec}} - \mathcal{W}_t^{\text{opt}}.$$

Theorem 4 (Inefficiency). *Under congestion externalities and symmetric agents,*

$$\Gamma_t > 0.$$

8 Efficiency Frontier

Definition 8. *A configuration lies on the efficiency frontier if no feasible change reduces both $\mathcal{W}_t$ and Γ_t simultaneously.*

9 Duality

Definition 9. *Let $\lambda_t(\kappa)$ and $\eta_t(\kappa)$ be Lagrange multipliers for congestion and capacity constraints.*

Theorem 5 (Dual representation). *On the efficiency frontier:*

$$\lambda_t(\kappa) = \eta_t(\kappa).$$

10 Mean-field Stability

Theorem 6. *A fixed point μ^* is locally stable if:*

$$\text{Re}(\lambda_i(D\mathcal{F})) < 0.$$

11 Conclusion

We present a stochastic routing and queueing model of institutional systems with endogenous congestion, mean-field dynamics, inefficiency relative to planner allocation, and a dual representation in terms of shadow prices. The framework provides a unified mathematical structure for analyzing institutional congestion across heterogeneous social domains.

The End